

Basic Information

Experiment No.: 06

Date: 24-09-2026

Title: Java NavigableMap Interface using TreeMap

Objective / Aim

To implement the NavigableMap interface using "TreeMap" and demonstrate navigation, entry retrieval, removal, and descending-order operations.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

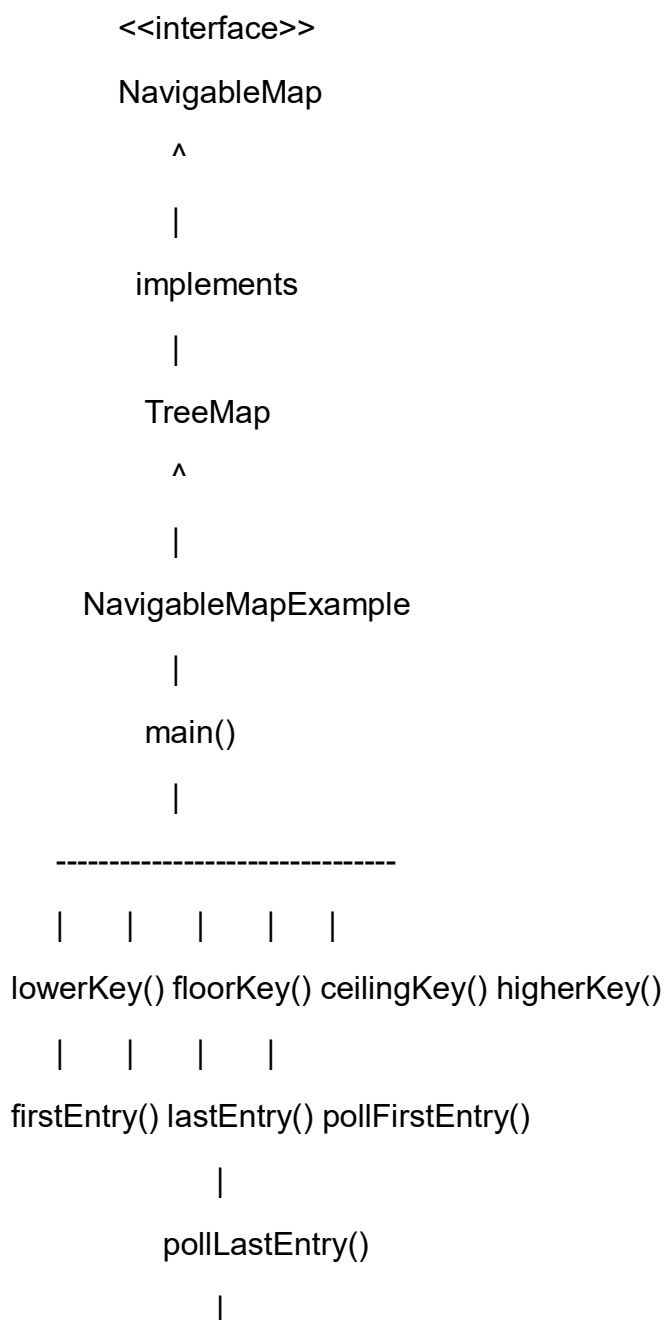
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "NavigableMap" using "TreeMap".
2. Add key-value pairs using "put()".
3. Find keys using "lowerKey()", "floorKey()", "ceilingKey()", and "higherKey()".
4. Display the first and last entries using "firstEntry()" and "lastEntry()".
5. Remove the first entry using "pollFirstEntry()".

6. Remove the last entry using "pollLastEntry()".
 7. Display the remaining map.
 8. Display the map in descending order using "descendingMap()".
 9. End the program.
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UML Class Diagram



descendingMap()

Source Code

```
package collectionframework;

import java.util.NavigableMap;
import java.util.TreeMap;

public class NavigableMapExample {
    public static void main(String[] args) {

        NavigableMap<Integer, String> map = new TreeMap<>();

        map.put(10, "A");
        map.put(20, "B");
        map.put(30, "C");
        map.put(40, "D");
        map.put(50, "E");

        System.out.println("Map: " + map);

        // lowerKey()
        System.out.println("Lower Key than 30: " +
            map.lowerKey(30));

        // floorKey()
        System.out.println("Floor Key of 30: " +
            map.floorKey(30));
```

```
// ceilingKey()
System.out.println("Ceiling Key of 25: " +
    map.ceilingKey(25));

// higherKey()
System.out.println("Higher Key than 30: " +
    map.higherKey(30));

// firstEntry()
System.out.println("First Entry: " +
    map.firstEntry());

// lastEntry()
System.out.println("Last Entry: " +
    map.lastEntry());

// pollFirstEntry()
System.out.println("Removed First Entry: " +
    map.pollFirstEntry());

// pollLastEntry()
System.out.println("Removed Last Entry: " +
    map.pollLastEntry());

System.out.println("After polling: " + map);

// descendingMap()
System.out.println("Descending Map: " +
```

```
        map.descendingMap());  
    }  
}
```

Compilation & Execution Commands

Save the file as:

NavigableMapExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

```
javac NavigableMapExample.java
```

Run:

```
java collectionframework.NavigableMapExample
```

Output Screen

Map: {10=A, 20=B, 30=C, 40=D, 50=E}

Lower Key than 30: 20

Floor Key of 30: 30

Ceiling Key of 25: 30

Higher Key than 30: 40

First Entry: 10=A

Last Entry: 50=E

Removed First Entry: 10=A

Removed Last Entry: 50=E

After polling: {20=B, 30=C, 40=D}

Descending Map: {40=D, 30=C, 20=B}

Result

The NavigableMap interface was successfully implemented using "TreeMap", and navigation, entry retrieval, polling, and descending-order operations were successfully demonstrated.